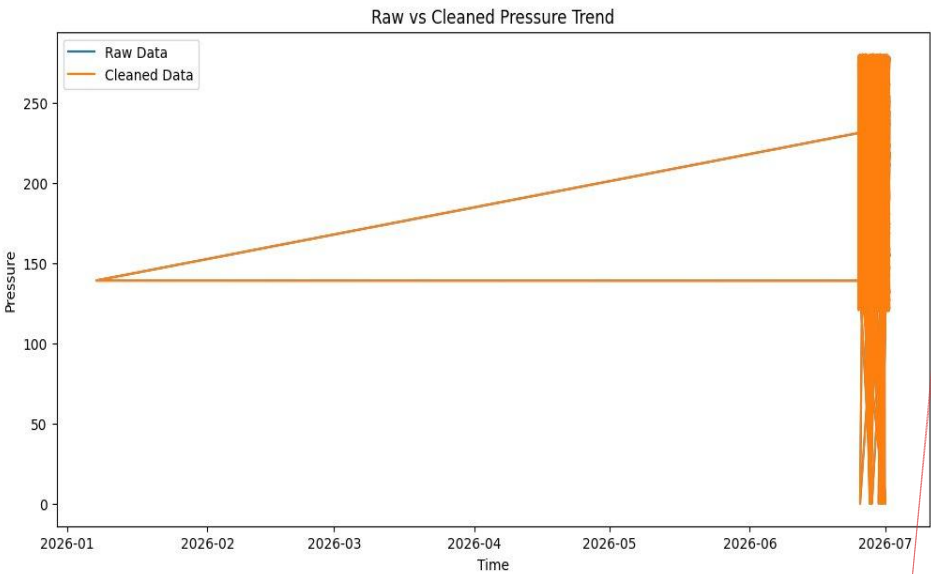


Week2_Insight_Report

Bramuel Wesonga | Raw vs. Cleaned Pressure Trend Analysis

Context – Insight – Action



Context: What was wrong with the raw data? The raw sensor dataset from the processing plant was unreliable. It contained missing pressure readings, duplicate records, inconsistent zone labels, and physically impossible values outside the 0-300 PSI operating range. The most critical problem appeared in late June 2026, where the raw data showed extreme pressure oscillations swinging between 0 and 280 PSI. Any operator reviewing this raw data would have concluded the plant was experiencing dangerous, violent pressure swings requiring immediate emergency action.

Insight: What hidden trend was revealed after cleaning? After cleaning, which involved interpolating missing values, removing duplicates, standardizing zone names, and filtering impossible sensor readings, a completely different story emerged. The raw data suggested violent

instability, but the cleaning process revealed a gradual, controlled pressure increase from approximately 140 PSI in January to 230 PSI by late June 2026. The extreme oscillations were not a real operational event. They were artifact noise from a failing pressure sensor, not an actual pressure surge. The hidden trend shows that the plant was executing a deliberate, planned pressure ramp-up over the six-month period, operating within safe parameters the entire time. The cleaning process did not just tidy the data; it uncovered the real operational picture.

Action: What operational change do you recommend? Replace the faulty pressure sensor in the affected zone immediately. This sensor is producing artifact noise that could trigger false alarms, cause unnecessary emergency shutdowns, or lead operators to make incorrect manual adjustments based on non-existent pressure events. Second, install a real-time data validation script that flags any reading outside the 0-300 PSI range and automatically interpolates short gaps under five minutes. Third, implement a weekly automated data quality report for the plant floor so sensor health issues are caught early, before they can mask real operational trends. Without these changes, the next sensor failure could erode operator trust in the data and lead to decisions based on noise instead of reality.